

GTU CSE 241 / 501 Quiz

Std Code: **158**

Std Number:

2 1 0 1 0 4 0 0 4 0 1 7

```
#include<iostream>
#include<vector>
using namespace std;

double nextArea(const vector<shape*>&U){
    double tot_sqr=0.0;
    double other=0.0;
    if(U.empty())exit(1);

    for(int i=0; i<U.size(); i++){ /*normal loop
    St vector*temp=dynamic_cast<shape*>(U[i]);
    if(temp!=nullptr) tot_sqr+=U[i]>area();
    else other+=U[i]>area();
    }
    /* if there are getters, other solution is
    for(const auto& temp :U){
        if(temp->get_len()==temp->get_wid+h())
            tot_sqr+=temp->area();
        else other+=temp->area();
    }
    */
    return tot_sqr - other;
}
```

```

int main() {
    Vector<Shape*> vec(5); // at least five
    Circle cir(3.0);
    Square sq(3.0, 3.0);
    Rectangle rec(2.0, 3.0);
    Circle cir_2(5.0);
    Square sq_2(4.0, 4.0);

    vec[0] = &cir;
    vec[1] = &sq;
    vec[2] = &rec;
    vec[3] = &cir_2;
    vec[4] = &sq_2;

    // I could pushback also, but I chose that way.
    double result = NetArea(vec);
    cout << "The result is -> " << result << endl;
    return 0;
}

```